A Cognitive Swarm of Small Local Models: Cross-Critique Debate Meets a Truth Hierarchy to Beat a 27B Cloud Model on 8 GB of RAM

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Abstract

Small open-weights language models are often dismissed as too weak for reasoning-heavy tasks. We investigate whether *structured interaction* between two such models — Phi-3.5 Mini (3.8B) and Qwen 2.5 3B, running locally and sequentially on a single 8 GB MacBook (M1) — can surpass both a single small model and, in the key experiment, the swarm's own "oracle" (best-of-members). We report three contributions. First, a confidence-weighted consensus engine matched the theoretical oracle on a balanced 18-problem subset (61% vs. 61%), doubling the earlier heuristic (33%). Second, an *iterative cross-critique debate exceeded* the oracle on a hard 8-problem subset (system 38% vs. oracle 12%; candidate pool 62%), demonstrating emergent generation of answers no single member produces alone. Third, we show that for real learning, consensus is not the primary mechanism: a *truth hierarchy* — verified memory → code execution → deterministic calculation → corroborated retrieval → debate-as-fallback — lifted the weakest category (code output prediction) from 8% to 100% with zero model loads, and resolved 42% of a 36-problem set in 0.35 s deterministically. We close with honest limits and identify selection (not generation) as the remaining bottleneck.

1 Introduction

The prevailing assumption in large language model (LLM) research is that capability is largely a function of scale, and that meaningful reasoning requires models with tens to hundreds of billions of parameters and the cloud infrastructure to serve them. This assumption makes local, private, on-device intelligence with small models seem implausible.

We revisit this assumption. Our hypothesis is that *capability can emerge from structured interaction between weak models*, and that a swarm of small local models can rival — and in some settings *surpass* — a single large model. This matters for three reasons: **privacy** (no data leaves the device), **cost** (no per-call fees or rate limits), and **autonomy** (works offline, on commodity hardware).

We constrain the problem severely: two small models (3.8B and 3B parameters) on a single 8 GB MacBook (M1), loading one model at a time. Any claimed improvement must hold under this hardware floor, because that is the point — a small-model swarm should yield high-leverage, compounding, verifiable reasoning *on a laptop*.

Contributions.

1. A **confidence-weighted consensus engine** that reaches the theoretical oracle ceiling on a balanced subset (61% vs. 61%), up from 33%

with a heuristic baseline, by fixing the root cause of low scores — verbose free-text answer extraction from local models.

2. An **iterative cross-critique debate** that *beats its own oracle*: on a hard 8-problem subset it generates correct answers (system 38%, candidate pool 62%) where the best single member achieves 12%. This is measurable emergence, not averaging.

3. A **truth hierarchy** that reframes consensus as a *fallback* rather than the primary mechanism: deterministic, corroborated sources of truth (verified memory, code execution, calculation, corroboration) outperform consensus everywhere they apply, lifting the weakest category from 8% to 100% with zero model loads.

2 Related Work

Multi-agent LLM debate. Several works propose having multiple model instances debate to improve factual consistency and reasoning quality [1, 2, 3]. These predominantly use large or frontier models and show the debate recovers individual-model capability. Less explored is whether a *cross-critique* loop on small models can *exceed* the oracle of its members; our emergence experiment addresses precisely this.

Retrieval-augmented generation (RAG). Grounding model outputs in external sources is established [4]. Our contribution is the *verified memory* with provenance, and the explicit truth-tier routing that prefers deterministic tools and honest conflict reporting over synthesis for ambiguous domains [5].

Model collapse. Self-training on model-generated data degrades distribution over generations [6]. Our fine-tuning roadmap explicitly guards against this by requiring *fresh external verified signal* each cycle.

Collaborative and local inference. Sequential loading to fit small models on constrained hardware, and prompt optimization to extract clean answers from terse local outputs, follow the general thrust of making LLMs usable on-device [7].

Table 1: Models and hardware.

Model	Params	Q4 RAM	Role
Phi-3.5 Mini	3.8B	∼2.5 GB	reasoning/math
Qwen 2.5 3B	3B	∼2.0 GB	math/code

Host: Apple MacBook (M1, 8 GB RAM). One model loaded at a time (MLX 4-bit). Qwen 27B (Groq API) used as a cloud comparison.

3 Method

3.1 Models and Hardware

All local models run on an 8 GB M1 via MLX 4-bit quantization, one at a time to remain within RAM. A large cloud model (Qwen 27B) serves as an external comparison point.

3.2 Answer Extraction

Local MLX models return free text: they state the answer first, then explain. Verbose output shattered regex extraction and dominated results. Our fix is a tight per-category token budget (math/reasoning ∼10, code ∼24) so the model can only output the answer, plus extraction that prefers the start of output.

3.3 Debate Protocols

Single-pass consensus (v3). Two first-guesses are aggregated by a deterministic engine [8] that clusters votes by answer, then scores each cluster by agreement, self-consistency, and confidence. Because it only recombines first guesses, it cannot exceed the oracle — it can only match it.

Iterative cross-critique (v4). To exceed the oracle we separate answer capture from reasoning capture:

- **Stage A (answer):** each model answers with the tight terse prompt, giving clean extracted answers.
- **Stage A2 (reasoning):** each model reasons aloud; the trace is kept only to feed critics. (Small models answer first, then explain.)
- **Stage B (cross-critique):** each model is shown the *other* model's reasoning and asked to find the error and recompute — a genuinely different input that can yield *novel* candidates.
- **Stage C (consensus):** all candidates are pooled and aggregated.

Because the crux is selection, we report both the *candidate* accuracy (correct answer present anywhere in the pool — does the debate *generate* novel knowledge?) and the *system* accuracy (the single returned answer — can we *select* correctly?).

3.4 The Truth Hierarchy

Consensus is a fallback, not the primary mechanism. Real learning uses the cheapest, most-certain source of truth first:

1. **Tier 0 — Memory:** already-verified answers with provenance {answer, confidence, tier, reliability, sources, disagreement}. Instant, no compute; experience compounds.
2. **Tier 1 — Code execution:** run the interpreter for code-output questions (ground truth).
3. **Tier 2 — Calculation:** deterministic evaluation of basic arithmetic.
4. **Tier 3 — Retrieval corroboration:** reliability-weighted, independence-penalized agreement across sources; surface *conflict* honestly (no forced single answer for contested domains).
5. **Tier 4 — Debate:** weak-model debate as a hypothesis source.

Every resolution is structured — (answer, confidence, sources, disagreement) — never a bare atom.

Table 2: Overall accuracy, 36-problem benchmark. Highest in each column in bold.

Approach	Overall	Easy	Med	Hard	Math	Reas	C
Phi-3.5 Mini (single)	28	67	8	8	42	25	
Qwen 2.5 3B (single)	28	33	42	8	17	33	
Debate (swarm)	**39**	42	33	**42**	**67**	**42**	
API (Qwen 27B)	8	8	8	8	8	0	

Values are percentages. "Reas"=reasoning.

4 Experiments

4.1 Benchmark

A 36-problem benchmark spans three difficulty levels (12 easy / 12 medium / 12 hard) and three categories (math / reasoning / code). Accuracy = problem solved if the answer is correct. Single runs are noisy, so we report "solved in any run" unless stated.

4.2 Leveled Benchmark (36 problems)

The locally-running swarm of two small models outperformed a 24× larger cloud model (39% vs. 8%) and was the strongest performer on the hardest problems (42% vs. 8% for any single two small models and vs.$\tilde{8}$% for the large model). The debate solved 7 problems that neither single small model could solve, evidencing complementary value rather than averaging.

Honest caveat: the large model was crippled in practice by a persistent "thinking-mode" that dumps reasoning text into the output instead of a clean answer, regardless of prompting. This is a real observed operational failure of that model in this usable pipeline, not a theoretical superiority claim.

4.3 Consensus reaches the oracle (v3)

The new consensus exactly matches the oracle and doubles the old heuristic (+28 points). Every problem either single model can solve, the swarm now solves. Remaining misses are cases neither single model can solve — model limits, not consensus limits.

Table 3: Consensus on a balanced 18-problem subset.

Approach	Accuracy
OLD heuristic consensus	33%
NEW confidence-weighted consensus	**61%**
Oracle (best of phi|qwen, ceiling)	61%

Table 4: Hard 8-problem subset (h-r-1, h-r-3, h-r-4, h-m-1, h-m-3, h-m-4, m-r-4, m-r-3).

Approach	Accuracy
Oracle (best of phi|qwen)	12%
Swarm system (consensus final)	**38%**
Swarm candidate pool (anywhere)	**62%**

4.4 Emergence: the swarm exceeds the oracle (v4)

The cross-critique debate *beat the oracle*: system-level 38% vs. 12% (3×); candidate pool 62% vs. 12% (5×). Two findings stand out:

1. **Generation works:** the debate *generates* answers neither member produces alone. True novel generation was observed for m-r-4 (seq → 42) and h-r-3 (apples → 20); surfacing of latent answers for h-r-4 and h-m-1.

2. **Selection is the new bottleneck:** the candidate pool (62%) is nearly 2× the system answer (38%); majority-weighted consensus discards novel-correct minority answers.

4.5 The truth hierarchy (v5)

The weakest category (code, 8%) became the strongest (100%) simply by *running* the code instead of trusting models to predict output. Memory archives every verified answer with provenance; repeats hit memory in milliseconds with zero inference, so experience compounds independently of any single model's weights. Memory lookup is *code-aware* to prevent distinct code questions from blurring together via token overlap.

Table 5: Deterministic tiers; all run with zero model loads.

Tier	Result
Code executor (12 code problems)	12/12 = **100%** (was 8%)
Calculator (basic arithmetic)	correct where applicable
Memory compounding (pass 2 of 36)	15 hits in 5.3 ms
Total deterministic resolve (36)	**15/36 = 42% in 0.35 s**

Honest conflict handling. For contested domains (e.g. history): two independent claims that converge yield a corroborated answer (Fall of Rome → 476 CE, confidence 0.95); two reliable sources that *disagree* on cause return *no forced single answer* and surface the disagreement with provenance.

5 Discussion

Three results together reframe the claim "small models are too weak." First, prompt structure + extraction recovers much of what is credited to scale (§3). Second, cross-critique debate produces genuinely *new* reasoning above the oracle (§4) — the strongest evidence that a swarm is more than the union of its members. Third, for real learning, consensus is not the answer: deterministic, corroborated ground truth dominates (§5).

The persistent bottleneck is **selection**. Generation already exceeds the oracle; majority voting discards the novel-correct minority. We believe selecting a confident, well-argued minority correction over a stable-but-wrong majority is the highest-leverage next step.

6 Conclusion and Future Work

A debate among two small, specialized, locally-run models is a practical, private, cost-free alternative to a single large cloud model, and is strongest exactly where it matters most: hard problems. Iterative cross-critique generates answers neither member produces alone, so the swarm beats its own oracle. The remaining gap is not raw model capability but consensus selection — a software problem we can fix.

Future work: (i) replace majority selection with a selector that trusts a confident minority correction; (ii) generalize the emergence experiment to the full 36-problem set; (iii) extend the connector layer (retrieval, databases, academic sources) behind the truth-hierarchy router; (iv) a growing-ceiling trainer that consumes fresh verified signal into verified memory (instant) and optionally into slow LoRA fine-tuning, guarding against model collapse; and (v) treat a multimodal model as the final-model target, reading far more source kinds while memory keeps knowledge current without growing the model.

Honest Limits

- **API artifact:** the large-model comparisons are confounded by an operational "thinking-mode" failure, not a clean theoretical superiority claim.
- **Small experimental scale:** 36 problems, single-run non-determinism; the emergence result is on an 8-problem hard subset.
- **Model limits remain:** hard percent-change (h-m-3), GCD (h-m-4) and jugs (m-r-3) stay unsolved by the whole pipeline.
- **8 GB is a hard model floor** per base model; the system ceiling grows through memory/connectors, the neuron ceiling is bounded by the chip.

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